

Aaina Garg

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EDUCATION

University of Toronto

Honours Bachelor of Science, Specialist in Computer Science(Focus in AI), Major in Economics
Rotman Commerce Certificate in Business Fundamentals

Expected : Spring 2027
Toronto, ON

- Distinctions:**
- **1 of 30** selected for **Technology Leadership Initiative** from **1000+ applicants** at the University of Toronto
 - **1 of 75** selected **nationally** for **AI4Good Machine Learning Fellowship, Google Deepmind**
 - **1 of 35** selected **nationally** for **Let's Solve it Fellowship** from **RBC Borealis**
 - **1 of 190** selected **globally** for the **Sentient Futures Fall 2026**

Club Leadership: Director of Technology, **UofT Open Source Intelligence** | Founder & President, **UofT Pilates & Yoga Club** | Consultant, **UofT 180 DC** | ML Associate, **UTMIST** | Conference Associate(x3), **UofT AI**

TECHNICAL SKILLS

Skills: Python, TypeScript, SQL, PyTorch, Hugging Face, RoBERTa, YOLOv5, LPRNet, MCP, Random Forest, Next.js, React, REST APIs, PostgreSQL, Docker, Microsoft Copilot Studio, Power Automate, NumPy, Pandas, Matplotlib, LiDAR, Cyclomedia, CROW

EXPERIENCE

Machine Learning Engineer - 1st Canadian member on the AI, Data team

May 2024 – Apr 2027

Arcadis [Term 3: Jan 2026 – April 2027 (continued) | Term 2: May – Aug 2025 | Term 1: May – Oct 2024]

Toronto, ON

- Productionized an **AI intake tool** using **A2A-compatible multi-agent orchestration** in **Microsoft Copilot Studio**, with **Power Automate** workflows, a **Next.js/React** front end, API integrations, and caching; standardized AI-project evaluations for **60K employees**, improving intake consistency by **60%** and reducing early-stage evaluation time by **70%**.
- Engineered and deployed an end-to-end **YOLOv5 + LPRNet** pipeline for automated vehicle/license-plate detection and recognition, including image ingestion, inference, confidence-based validation, and results logging; achieved **96%+** detection accuracy across **1000s of concurrent requests**, reduced manual validation by **60%**, saved **\$50K**, and strengthened a client-facing solution offering.
- Architected a road-marking quality pipeline with **Geomaat LiDAR**, marking-location data, and **Cyclomedia** detections; engineered data-quality checks and generated **CROW** condition scores for maintenance prioritization, trained random forest model for accuracy.

AI Governance Research Fellow

Aug 2026 – Present

Sentient Futures Project Incubator

Remote

- Proposed and am building a "**Sentience-Aware LLM Evaluation Toolkit**" testing how frontier LLMs reason about **artificial sentience** and **moral status under adversarial pressure**, shipping both the evaluation harness (**code + simple interface**) and a **public write-up**, so digital-minds and AI governance researchers can run it directly.
- Building scenario sets targeting three specific failure modes: **inconsistent judgments across structurally similar cases, denial of moral status** under strong evidence, and welfare-blind reasoning in agentic decision tasks : selected as **1 of 190 mentees globally**.

AI Research Intern

Apr 2026 – Present

Cohuman Labs

Toronto, ON

- Co-developing a **Return-on-Value-of-AI** framework quantifying **EU AI Act** compliance economics covering financial exposure and reputational value erosion via event-study methodology, and agentic-AI accountability gaps for a white paper & an academic paper.

Applied ML Research Fellow

Jun 2025 – Sep 2025

Google DeepMind, Vector Institute, MILA, CIFAR via **AI4Good Fellowship**

Toronto, ON

- Developed a bias-detection system using **RoBERTa** classifier and applied **adversarial debiasing** (discriminator penalizing protected-attribute leakage) to reduce the gender true-positive-rate gap by **68%** with under **2%** accuracy loss.
- Built an interactive React interface enabling real-time bias analytics and audit trails, improving **interpretability** for non-technical users by **47%** while implementing responsive design patterns and client-side caching.
- Architected a **scalable** backend with **PostgreSQL database design, RESTful API endpoints, and Docker containerization**, supporting concurrent user sessions with proper error handling and comprehensive logging.

Software and Machine Learning Engineering Fellow

April 2026 – July 2026

RBC Borealis, via **Let's Solve it Fellowship**

Toronto, ON

- **Designed and deployed** an accessible, AI-enabled course-planning and student-progress web dashboard (**Claude Code and Google Antigravity**) for **Canadian National Institute of Blind Foundation's educators with low vision**, centralizing individualized learning plans, progress insights, and administrative workflows; **estimated to save 50+ educator hours per week**.

Software Engineer Apprentice

Sept 2024 – Dec 2024

Cash App [Square], via **UofT Technology Leadership Initiative**

Toronto, ON

- Built an interactive **bias-analytics dashboard** using **Python, NumPy, Pandas, Matplotlib, TypeScript, CSS**; partnered with ML engineers to visualize performance disparities, dynamic filters, and drill-down analyses for potential fairness vulnerabilities in model outputs.
- Implemented client-side aggregation and caching for responsive, audit-ready bias metrics, reducing server load by **60%**; translated technical fairness signals into actionable visualizations for both technical and non-technical stakeholders.

Product Strategy Consultant

Sept 2026-Dec 2026

Embark via **UofT Technology Leadership Initiative**

Toronto, ON

- Designed an **LLM-powered student-planning hub** that aggregated university, course, residence, club, and research information across **23 Ontario universities**, creating value for guidance counsellors and trusted planning support for students and parents.
- Built an **8-week pilot** and revenue model targeting **10 counsellors** and **4,000 students**, linking late-high-school engagement to Embark's RESP conversion funnel and projecting **\$4.4M AUM** and **\$28.1K** net earnings.